Compare the flow through two routes

Task 3.6–3.7 / Prepare and draw

Name _____ Team _____ Date _____

01 Set up the investigation

Sketch routes A and B. Change one variable, such as tube length. Use the same water level or drive. Flow rate = collected volume ÷ collection time.

Mark a common head or driving condition. Measure tube length, collection time, and collected volume.

Route	Tube length / bore	Drive condition	Fixed outlet position
A			
B			

02 Plan it on paper

Draw two routes differing in one variable. Predict which will deliver more water in equal time.

My prediction, chosen feature or design limit

Compare the flow through two routes

Task 3.6–3.7 / Test, compare and explain

Name _____ Team _____ Date _____

03 Run the test and record what happened

Run both routes from the same starting condition. Keep tube ends and water level consistent. Collect three trials. Calculate volume/time with units. Check for kinks or changing head. Compare the direction of an outlet jet with the expected reaction.

Trial	A volume / time	B volume / time	A and B flow rates with units
1			
2			
3			

04 Use the reading

Ch. 22 | Good Chemistry | printed pp. 199-207
Ch. 23 | Down the Pipes | printed pp. 209-218
Complete the assigned checks in the interactive reader. Here, identify a page and one idea that helps explain your result or design decision.

05 Individual explanation

Which change would you recommend testing on the boat or press? Support your prediction with measured flow rates.

Keep these sheets with the work. Save the individual explanation in the lesson’s Explain step.